

Christian_Herrera_HW10

April 7, 2026

1 Homework10- Advance Pandas (Questions)

2 Assignment Submission Guidelines

Please follow the guidelines below for submitting your assignment:

1. Submission Deadline:

- All assignments must be submitted **no later than 23:59 PM next Thursday (04/09)**.
- Late submissions will not be accepted unless prior arrangements have been made by the TAs.

2. Complete Your Work:

- Finish your work in Google Colab and ensure that your notebook is saved.

3. Submit Your Assignment on Canvas:

- Log in to **Canvas** and navigate to the assignment submission page.

4. File Naming Convention:

- Please name your files as follows: `Lastname_Firstname_AssignmentName`
- Example: `Alex_John_HW10.ipynb`, `Alex_John_HW10.pdf`

5. Technical Issues:

- If you encounter any technical issues with Canvas or your submission, please contact the TAs immediately **before the deadline** to avoid penalties.

3 Questions

1. Load the Startups dataset into a pandas DataFrame and display the first few rows. (5 Points)

```
[39]: import pandas as pd
df = pd.read_csv('startups.csv')

df.head()
```

```
[39]:   R&D Spend  Administration  Marketing Spend   State  Profit
0  165349.20      136897.80      471784.10  New York  192261.83
```

1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

2. Find the number of rows and columns in the dataset. (5 Points)

```
[40]: print(f'rows: {df.shape[0]} cols: {df.shape[1]}')
```

```
rows: 50 cols: 5
```

3. Check for missing values in each column and handle them appropriately. (5 Points)

```
[41]: #df['R&D Spend'].isna().sum()

print(df.isna().sum())
```

```
R&D Spend      0
Administration 0
Marketing Spend 0
State           0
Profit          0
dtype: int64
```

4. Calculate the mean, median, and standard deviation of the 'R&D_Spend' column. (5 Points)

```
[42]: print('general stats:\n' , df['R&D Spend'].describe(), '\n') # The shortcut

# One by one
print('mean:', df['R&D Spend'].mean(),
      'median:', df['R&D Spend'].median(),
      'std:', df['R&D Spend'].std())
```

```
general stats:
count      50.000000
mean       73721.615600
std        45902.256482
min         0.000000
25%        39936.370000
50%        73051.080000
75%        101602.800000
```

```
max      165349.200000
Name: R&D Spend, dtype: float64
```

```
mean: 73721.6156 median: 73051.08 std: 45902.25648230753
```

5. Filter and create a new DataFrame containing companies located in a **California**. (5 Points)

```
[43]: cali_df = df[df['State'] == 'California']
      cali_df
```

```
[43]:
```

	R&D Spend	Administration	Marketing Spend	State	Profit
1	162597.70	151377.59	443898.53	California	191792.06
6	134615.46	147198.87	127716.82	California	156122.51
9	123334.88	108679.17	304981.62	California	149759.96
11	100671.96	91790.61	249744.55	California	144259.40
13	91992.39	135495.07	252664.93	California	134307.35
16	78013.11	121597.55	264346.06	California	126992.93
20	76253.86	113867.30	298664.47	California	118474.03
25	64664.71	139553.16	137962.62	California	107404.34
32	63408.86	129219.61	46085.25	California	97427.84
34	46426.07	157693.92	210797.67	California	96712.80
37	44069.95	51283.14	197029.42	California	89949.14
39	38558.51	82982.09	174999.30	California	81005.76
40	28754.33	118546.05	172795.67	California	78239.91
42	23640.93	96189.63	148001.11	California	71498.49
44	22177.74	154806.14	28334.72	California	65200.33
47	0.00	135426.92	0.00	California	42559.73
49	0.00	116983.80	45173.06	California	14681.40

6. Sort the dataset by 'Profit' in descending order. (5 Points)

```
[44]: sort_by_profit_df = df.sort_values('Profit', ascending=False)
      sort_by_profit_df
```

```
[44]:
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94
5	131876.90	99814.71	362861.36	New York	156991.12
6	134615.46	147198.87	127716.82	California	156122.51
7	130298.13	145530.06	323876.68	Florida	155752.60

8	120542.52	148718.95	311613.29	New York	152211.77
9	123334.88	108679.17	304981.62	California	149759.96
10	101913.08	110594.11	229160.95	Florida	146121.95
11	100671.96	91790.61	249744.55	California	144259.40
12	93863.75	127320.38	249839.44	Florida	141585.52
13	91992.39	135495.07	252664.93	California	134307.35
14	119943.24	156547.42	256512.92	Florida	132602.65
15	114523.61	122616.84	261776.23	New York	129917.04
16	78013.11	121597.55	264346.06	California	126992.93
17	94657.16	145077.58	282574.31	New York	125370.37
18	91749.16	114175.79	294919.57	Florida	124266.90
19	86419.70	153514.11	0.00	New York	122776.86
20	76253.86	113867.30	298664.47	California	118474.03
21	78389.47	153773.43	299737.29	New York	111313.02
22	73994.56	122782.75	303319.26	Florida	110352.25
23	67532.53	105751.03	304768.73	Florida	108733.99
24	77044.01	99281.34	140574.81	New York	108552.04
25	64664.71	139553.16	137962.62	California	107404.34
26	75328.87	144135.98	134050.07	Florida	105733.54
27	72107.60	127864.55	353183.81	New York	105008.31
28	66051.52	182645.56	118148.20	Florida	103282.38
29	65605.48	153032.06	107138.38	New York	101004.64
30	61994.48	115641.28	91131.24	Florida	99937.59
31	61136.38	152701.92	88218.23	New York	97483.56
32	63408.86	129219.61	46085.25	California	97427.84
33	55493.95	103057.49	214634.81	Florida	96778.92
34	46426.07	157693.92	210797.67	California	96712.80
35	46014.02	85047.44	205517.64	New York	96479.51
36	28663.76	127056.21	201126.82	Florida	90708.19
37	44069.95	51283.14	197029.42	California	89949.14
38	20229.59	65947.93	185265.10	New York	81229.06
39	38558.51	82982.09	174999.30	California	81005.76
40	28754.33	118546.05	172795.67	California	78239.91
41	27892.92	84710.77	164470.71	Florida	77798.83
42	23640.93	96189.63	148001.11	California	71498.49
43	15505.73	127382.30	35534.17	New York	69758.98
44	22177.74	154806.14	28334.72	California	65200.33
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116983.80	45173.06	California	14681.40

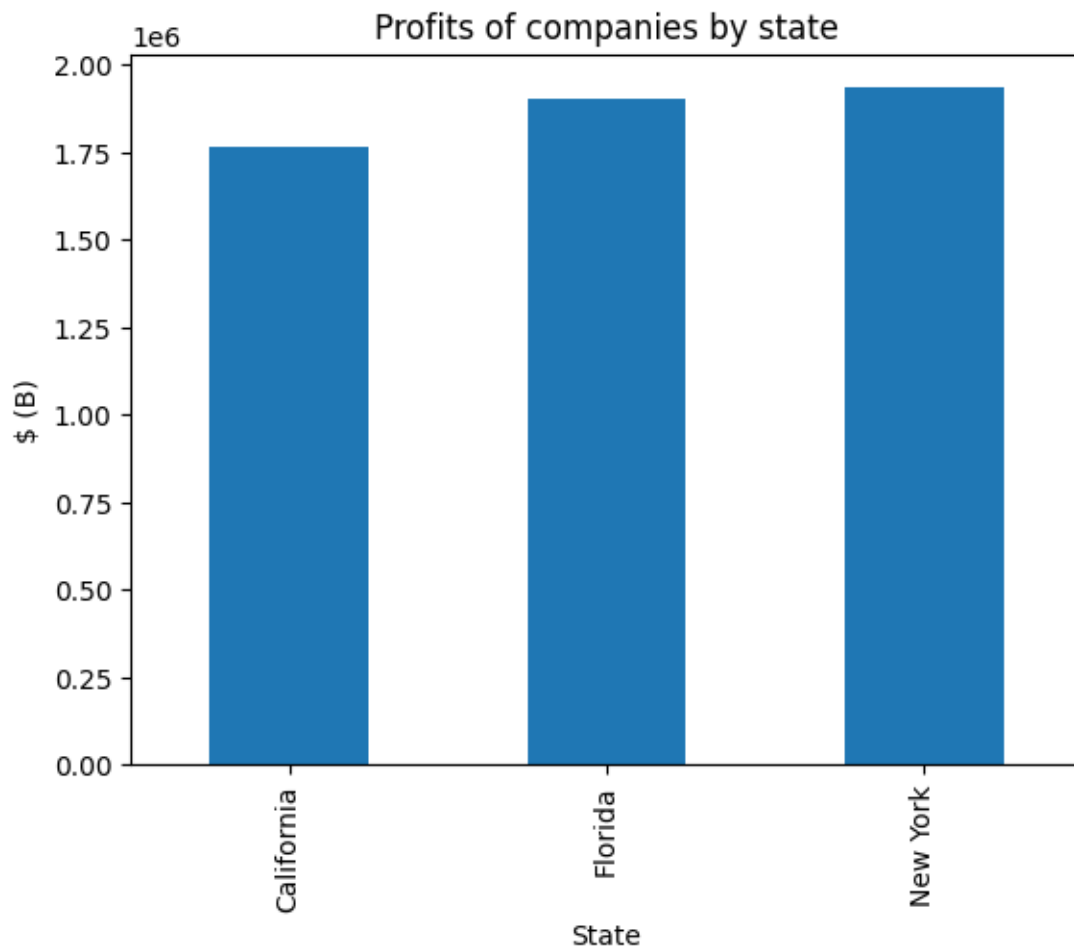
7. Group the data by 'State' and calculate the total 'Profit' for each state. (5 Points)

```
[45]: state_profit_df = df.groupby('State')['Profit'].sum()
state_profit_df
```

```
[45]: State
California    1766387.98
Florida       1900384.39
New York      1933859.59
Name: Profit, dtype: float64
```

8. Create a bar chart to visualize the total 'Profit' by 'State.' (5 Points)

```
[46]: import matplotlib.pyplot as plt
state_profit_df.plot(kind='bar')
plt.title('Profits of companies by state')
plt.ylabel('$ (B)') # I'm just guessing that this is in billions.
plt.xlabel('State')
plt.show()
```



9. Calculate the correlation between 'R&D_Spend' and 'Profit.' (5 Points)

```
[47]: correlation = df['R&D Spend'].corr(df['Profit'])
      print(f"Correlation between R&D Spend and Profit: {correlation:.4f}")
```

Correlation between R&D Spend and Profit: 0.9729

10. Create a new column 'Total_Spend' that represents the sum of 'R&D_Spend,' 'Administration,' and 'Marketing_Spend' for each company. (5 Points)

```
[48]: df['Total_Spend'] = df[['R&D Spend', 'Administration', 'Marketing Spend']].
      ↪sum(axis=1)
      df
```

```
[48]:
```

	R&D Spend	Administration	Marketing Spend	State	Profit \
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
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5	131876.90	99814.71	362861.36	New York	156991.12
6	134615.46	147198.87	127716.82	California	156122.51
7	130298.13	145530.06	323876.68	Florida	155752.60
8	120542.52	148718.95	311613.29	New York	152211.77
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22	73994.56	122782.75	303319.26	Florida	110352.25

23	67532.53	105751.03	304768.73	Florida	108733.99
24	77044.01	99281.34	140574.81	New York	108552.04
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26	75328.87	144135.98	134050.07	Florida	105733.54
27	72107.60	127864.55	353183.81	New York	105008.31
28	66051.52	182645.56	118148.20	Florida	103282.38
29	65605.48	153032.06	107138.38	New York	101004.64
30	61994.48	115641.28	91131.24	Florida	99937.59
31	61136.38	152701.92	88218.23	New York	97483.56
32	63408.86	129219.61	46085.25	California	97427.84
33	55493.95	103057.49	214634.81	Florida	96778.92
34	46426.07	157693.92	210797.67	California	96712.80
35	46014.02	85047.44	205517.64	New York	96479.51
36	28663.76	127056.21	201126.82	Florida	90708.19
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41	27892.92	84710.77	164470.71	Florida	77798.83
42	23640.93	96189.63	148001.11	California	71498.49
43	15505.73	127382.30	35534.17	New York	69758.98
44	22177.74	154806.14	28334.72	California	65200.33
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116983.80	45173.06	California	14681.40

	Total_Spend
0	774031.10
1	757873.82
2	662521.60
3	646243.88
4	599667.53
5	594552.97
6	409531.15
7	599704.87
8	580874.76
9	536995.67
10	441668.14
11	442207.12
12	471023.57
13	480152.39
14	533003.58
15	498916.68
16	463956.72
17	522309.05

```

18    500844.52
19    239933.81
20    488785.63
21    531900.19
22    500096.57
23    478052.29
24    316900.16
25    342180.49
26    353514.92
27    553155.96
28    366845.28
29    325775.92
30    268767.00
31    302056.53
32    238713.72
33    373186.25
34    414917.66
35    336579.10
36    356846.79
37    292382.51
38    271442.62
39    296539.90
40    320096.05
41    277074.40
42    267831.67
43    178422.20
44    205318.60
45    127057.20
46    414246.13
47    135426.92
48     52285.20
49    162156.86

```

11. Calculate the average **R&D_Spend**, **Administration**, and **Marketing_Spend** for companies located in each **State** (5 Points)

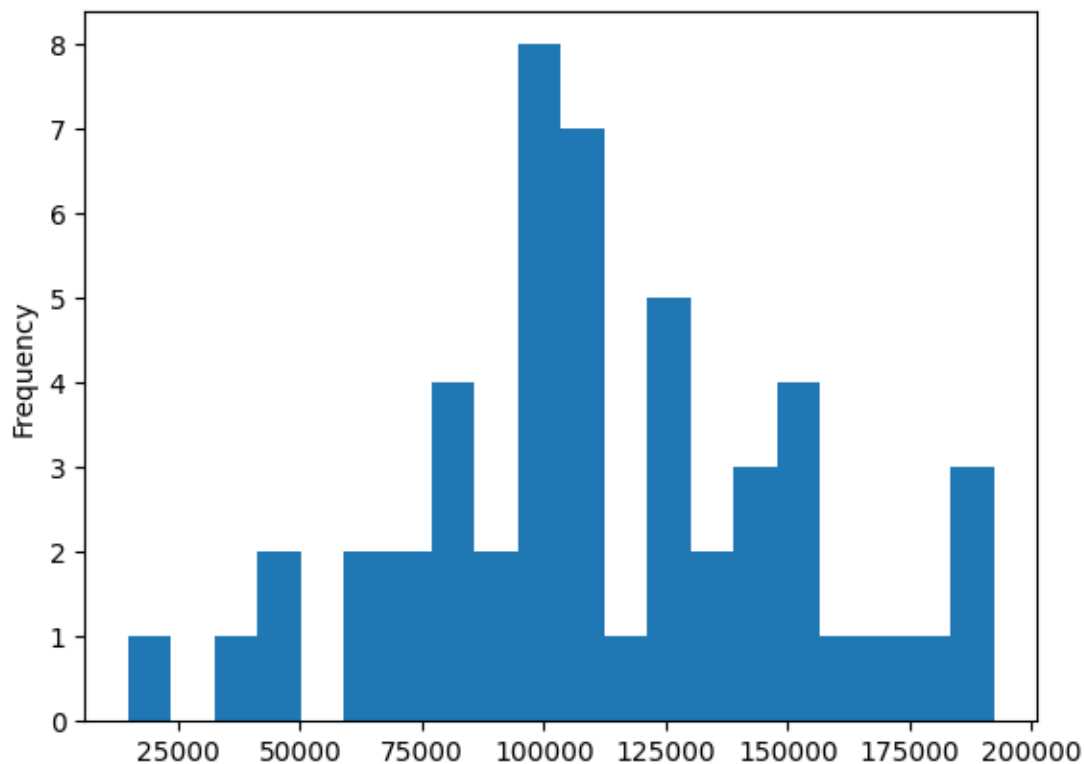
```
[49]: df.groupby('State')[['R&D Spend', 'Administration', 'Marketing Spend']].mean()
```

```
[49]:
```

	R&D Spend	Administration	Marketing Spend
State			
California	64657.674118	120746.507059	182540.929412
Florida	80724.016250	121768.897500	247323.551250
New York	76195.062353	121543.470588	205346.015882

12. Create a histogram to visualize the distribution of 'Profit.' (5 Points)

```
[50]: df['Profit'].plot(kind='hist', bins=20)
plt.show()
```



-
13. Filter the dataset to include only companies with 'R&D_Spend' greater than a 12000. (5 Points)

```
[51]: high_rd_df = df[df['R&D Spend'] > 12000]
high_rd_df.head() # Only the top 5.
```

```
[51]:
```

	R&D Spend	Administration	Marketing Spend	State	Profit	\
0	165349.20	136897.80	471784.10	New York	192261.83	
1	162597.70	151377.59	443898.53	California	191792.06	
2	153441.51	101145.55	407934.54	Florida	191050.39	
3	144372.41	118671.85	383199.62	New York	182901.99	
4	142107.34	91391.77	366168.42	Florida	166187.94	

	Total_Spend
0	774031.10
1	757873.82

```
2    662521.60
3    646243.88
4    599667.53
```

-
-
14. Calculate the total 'Profit' for companies with 'Marketing_Spend' above a certain threshold. (5 Points)

```
[58]: threshold = 156991 # Top 5 companies only.
print(df[df['Marketing Spend'] > threshold]['Profit'].sum())
```

```
4136608.24
```

15. Replace any missing values in the 'Marketing_Spend' column with the mean spending of the dataset. (5 Points)

```
[61]: df['Marketing Spend'] = df['Marketing Spend'].fillna(df['Marketing Spend'].
↳mean())
df.head()
```

```
[61]:   R&D Spend  Administration  Marketing Spend   State  Profit \
0  165349.20    136897.80    471784.10  New York  192261.83
1  162597.70    151377.59    443898.53  California  191792.06
2  153441.51    101145.55    407934.54   Florida  191050.39
3  144372.41    118671.85    383199.62  New York  182901.99
4  142107.34     91391.77    366168.42   Florida  166187.94
```

```
   Total_Spend  Profit_Margin
0    774031.10    0.248390
1    757873.82    0.253066
2    662521.60    0.288369
3    646243.88    0.283023
4    599667.53    0.277133
```

-
-
16. Create a new column 'Profit_Margin' that represents the profit margin for each company (Profit / Total_Spend). (5 Points)

```
[62]: df['Profit_Margin'] = df['Profit']/df['Total_Spend']

df.head()
```

```
[62]:   R&D Spend  Administration  Marketing Spend   State  Profit \
0  165349.20    136897.80    471784.10  New York  192261.83
1  162597.70    151377.59    443898.53  California  191792.06
```

2	153441.51	101145.55	407934.54	Florida	191050.39
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	Total_Spend	Profit_Margin
0	774031.10	0.248390
1	757873.82	0.253066
2	662521.60	0.288369
3	646243.88	0.283023
4	599667.53	0.277133

17. Sort the dataset by 'Profit_Margin' in descending order. (5 Points)

```
[55]: df.sort_values('Profit_Margin', ascending=False)
```

```
[55]:
```

	R&D Spend	Administration	Marketing Spend	State	Profit	\
48	542.05	51743.15	0.00	New York	35673.41	
19	86419.70	153514.11	0.00	New York	122776.86	
45	1000.23	124153.04	1903.93	New York	64926.08	
32	63408.86	129219.61	46085.25	California	97427.84	
43	15505.73	127382.30	35534.17	New York	69758.98	
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30	61994.48	115641.28	91131.24	Florida	99937.59	
24	77044.01	99281.34	140574.81	New York	108552.04	
10	101913.08	110594.11	229160.95	Florida	146121.95	
11	100671.96	91790.61	249744.55	California	144259.40	
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44	22177.74	154806.14	28334.72	California	65200.33	
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25	64664.71	139553.16	137962.62	California	107404.34	
29	65605.48	153032.06	107138.38	New York	101004.64	
37	44069.95	51283.14	197029.42	California	89949.14	
12	93863.75	127320.38	249839.44	Florida	141585.52	
38	20229.59	65947.93	185265.10	New York	81229.06	
26	75328.87	144135.98	134050.07	Florida	105733.54	
2	153441.51	101145.55	407934.54	Florida	191050.39	
35	46014.02	85047.44	205517.64	New York	96479.51	
3	144372.41	118671.85	383199.62	New York	182901.99	
28	66051.52	182645.56	118148.20	Florida	103282.38	
41	27892.92	84710.77	164470.71	Florida	77798.83	
13	91992.39	135495.07	252664.93	California	134307.35	
9	123334.88	108679.17	304981.62	California	149759.96	
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16	78013.11	121597.55	264346.06	California	126992.93
39	38558.51	82982.09	174999.30	California	81005.76
42	23640.93	96189.63	148001.11	California	71498.49
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8	120542.52	148718.95	311613.29	New York	152211.77
15	114523.61	122616.84	261776.23	New York	129917.04
7	130298.13	145530.06	323876.68	Florida	155752.60
33	55493.95	103057.49	214634.81	Florida	96778.92
36	28663.76	127056.21	201126.82	Florida	90708.19
1	162597.70	151377.59	443898.53	California	191792.06
14	119943.24	156547.42	256512.92	Florida	132602.65
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40	28754.33	118546.05	172795.67	California	78239.91
20	76253.86	113867.30	298664.47	California	118474.03
17	94657.16	145077.58	282574.31	New York	125370.37
34	46426.07	157693.92	210797.67	California	96712.80
23	67532.53	105751.03	304768.73	Florida	108733.99
22	73994.56	122782.75	303319.26	Florida	110352.25
21	78389.47	153773.43	299737.29	New York	111313.02
27	72107.60	127864.55	353183.81	New York	105008.31
46	1315.46	115816.21	297114.46	Florida	49490.75
49	0.00	116983.80	45173.06	California	14681.40

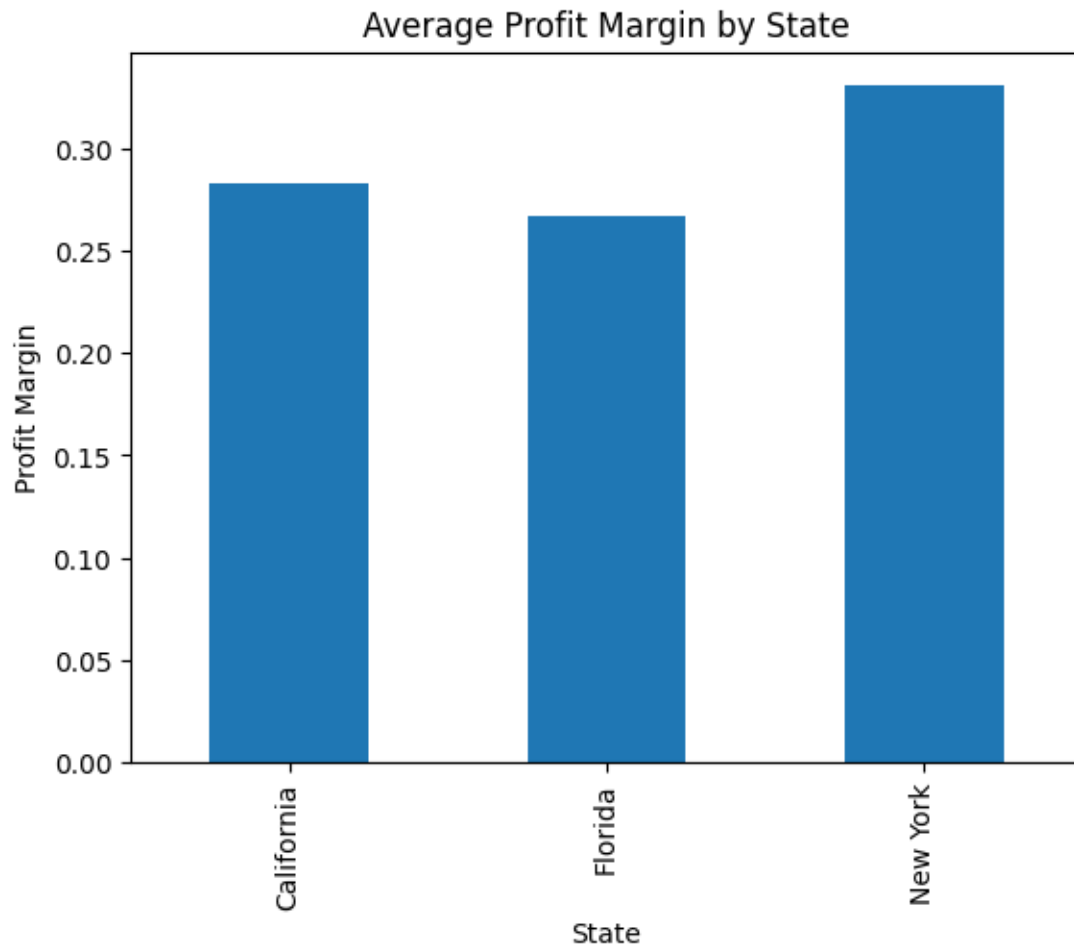
	Total_Spend	Profit_Margin
48	52285.20	0.682285
19	239933.81	0.511711
45	127057.20	0.510999
32	238713.72	0.408137
43	178422.20	0.390977
6	409531.15	0.381223
30	268767.00	0.371837
24	316900.16	0.342543
10	441668.14	0.330841
11	442207.12	0.326226
31	302056.53	0.322733
44	205318.60	0.317557
47	135426.92	0.314263
25	342180.49	0.313882
29	325775.92	0.310043
37	292382.51	0.307642
12	471023.57	0.300591
38	271442.62	0.299249
26	353514.92	0.299092
2	662521.60	0.288369
35	336579.10	0.286647
3	646243.88	0.283023

28	366845.28	0.281542
41	277074.40	0.280787
13	480152.39	0.279718
9	536995.67	0.278885
4	599667.53	0.277133
16	463956.72	0.273717
39	296539.90	0.273170
42	267831.67	0.266953
5	594552.97	0.264049
8	580874.76	0.262039
15	498916.68	0.260398
7	599704.87	0.259715
33	373186.25	0.259331
36	356846.79	0.254194
1	757873.82	0.253066
14	533003.58	0.248784
0	774031.10	0.248390
18	500844.52	0.248115
40	320096.05	0.244426
20	488785.63	0.242384
17	522309.05	0.240031
34	414917.66	0.233089
23	478052.29	0.227452
22	500096.57	0.220662
21	531900.19	0.209274
27	553155.96	0.189835
46	414246.13	0.119472
49	162156.86	0.090538

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18. Calculate the average profit margin for companies in each 'State' and create a bar chart to display it. (5 Points)

```
[56]: df.groupby('State')['Profit_Margin'].mean().plot(kind='bar')
plt.title('Average Profit Margin by State')
plt.ylabel('Profit Margin')
plt.xlabel('State')
```

```
[56]: Text(0.5, 0, 'State')
```



19. Find the maximum 'R&D_Spend' in each **State**. (5 Points)

```
[57]: df.groupby('State')['R&D Spend'].max()
```

```
[57]: State
California    162597.70
Florida       153441.51
New York      165349.20
Name: R&D Spend, dtype: float64
```